

Answers

Exercise 1 :

For each of the following statements, determine whether the statement is true or false and prove your answer.

1. For all integers a and b , if a and $a^2 - b^2$ are both odd then b is even.

2. For any rational numbers x, y, z and t , we have

$$x + y\sqrt{2} = z + t\sqrt{2} \iff (x = z \wedge y = t).$$

3. For any two sets A and B , $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$.

4. For any map $h : E \longrightarrow F$ and any $C \subseteq F$, if $C \neq \emptyset$, then $h^{-1}(C) \neq \emptyset$.

5. For any map $h : E \longrightarrow F$ and any $A \subseteq E$, we have $h^{-1}(h(A)) = A$.

6. The map $f : \mathbb{N}^* \longrightarrow \mathbb{Z}$ given by $f(n) := (-1)^n E(\frac{n}{2})$ is bijective.

7. The map $g : \mathbb{N} \longrightarrow \mathbb{N}$ given by $g(n) :=$ the sum of the digits of n isn't surjective.

Solution: $(1+2+1.5+2+2+2+1.5)$

1. True.

$$\begin{aligned} a \text{ and } (a^2 - b^2) \text{ are both odd} &\implies a^2 \text{ and } (a^2 - b^2) \text{ are both odd} \\ &\implies a^2 - (a^2 - b^2) = b^2 \text{ is even} \\ &\implies b \text{ is even.} \end{aligned}$$

2. True.

$\implies$) Assume that $y \neq t$. We have

$$\begin{aligned} x + y\sqrt{2} = z + t\sqrt{2} &\implies (y - t)\sqrt{2} = z - x \\ &\implies \sqrt{2} = \frac{z - x}{y - t} \in \mathbb{Q} \end{aligned}$$

which is impossible. Thus, $y = t$ and so $x = z$.

$\longleftarrow$)

$$\begin{aligned} (x = z \wedge y = t) &\implies (x = z \wedge \sqrt{2}y = \sqrt{2}t) \\ &\implies x + \sqrt{2}y = z + \sqrt{2}t. \end{aligned}$$

3. False.

If we take $A := \{1\}$ and $B := \{2\}$, then $A \cup B = \{1, 2\}$. We have

$$\mathcal{P}(A) = \{\emptyset, \{1\}\} \quad \text{and} \quad \mathcal{P}(B) = \{\emptyset, \{2\}\},$$

so

$$\mathcal{P}(A) \cup \mathcal{P}(B) = \{\emptyset, \{1\}, \{2\}\}.$$

But

$$\mathcal{P}(A \cup B) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

4. False.

Let consider the map h defined as

$$\begin{aligned} h : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^2. \end{aligned}$$

If we take $C := \{-1\}$, we have $C \neq \emptyset$ and $h^{-1}(C) = \{x \in \mathbb{R} : x^2 = -1\} = \emptyset$.

5. False.

Let consider the map h defined as

$$\begin{aligned} h : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^2. \end{aligned}$$

If we take $A := \{1\}$, we have $h^{-1}(h(\{1\})) = h^{-1}(\{1\}) = \{-1, 1\} \neq A$.

6. True.

We have

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even,} \\ -\frac{n-1}{2}, & \text{if } n \text{ is odd.} \end{cases}$$

Injectivity : Let $n, m \in \mathbb{N}^*$, such that $f(n) = f(m)$, then $(-1)^n E\left(\frac{n}{2}\right) = (-1)^m E\left(\frac{m}{2}\right)$, so $(-1)^{n-m} E\left(\frac{n}{2}\right) = E\left(\frac{m}{2}\right)$.

Since $\forall n \in \mathbb{N}^*, E\left(\frac{n}{2}\right) \geq 0$, then we deduce that n and m have the same parity. Thus,

- if n, m are even, then $f(n) = f(m) \implies \frac{n}{2} = \frac{m}{2} \implies n = m$;
- if n, m are odd, then $f(n) = f(m) \implies -\frac{n-1}{2} = -\frac{m-1}{2} \implies n = m$.

Therefore, f is injective.

Surjectivity : Let $a \in \mathbb{Z}$.

- If $a > 0$, then $\exists n := 2a \in \mathbb{N}^*$ such that

$$f(n) = (-1)^{2a} E\left(\frac{2a}{2}\right) = a.$$

- If $a \leq 0$, then $\exists n := 1 - 2a \in \mathbb{N}^*$ such that

$$f(n) = (-1)^{1-2a} E\left(\frac{1-2a}{2}\right) = -(-a) = a.$$

7. False.

If $m \in \mathbb{N}^*$, then $\exists n := \underbrace{11 \dots 1}_{m \text{ times}}$ such that $g(n) = m$.

If $m = 0$, then $\exists n := 0$ such that $g(n) = m$.

Exercise 2 :

Consider the map

$$f: \mathbb{Z} \setminus \{0\} \longrightarrow \mathbb{Q}$$

$$n \longmapsto 1 + \frac{2}{n} + \frac{3}{n^2}.$$

1. Determine $f^{-1}(\{7\})$. Is f surjective ?

2. Let $n, m \in \mathbb{Z}$. Show that

$$3n + 3m + 2nm = 0 \implies (n, m) \in \{(-6, -2), (-3, -3), (-2, -6), (-1, 3), (0, 0), (3, -1)\}.$$

3. Deduce that f isn't injective.

4. Find a maximal set¹ $A \subset \mathbb{Z}$ for which the map $f|_A$ is injective. Justify your answer.

5. Let $B := \{n \in \mathbb{Z} : |2n^2 + n - 4| \leq 2\}$. Determine $f(B)$ and $f^{-1}(B)$.

Solution: (1+2+1+2+2)

1. We have

$$\begin{aligned} f^{-1}(\{7\}) &= \{n \in \mathbb{Z} \setminus \{0\} : f(n) = 7\} \\ &= \{n \in \mathbb{Z} \setminus \{0\} : -6n^2 + 2n + 3 = 0\} \\ &= \emptyset. \end{aligned}$$

Or, for all $n \in \mathbb{Z} \setminus \{0\}$, we have

$$|f(n)| \leq 1 + \frac{2}{|n|} + \frac{3}{n^2} \leq 6.$$

Thus, the equation $f(n) = 7$ has no solutions in $\mathbb{Z} \setminus \{0\}$ and so $f^{-1}(\{7\}) = \emptyset$.

2. We have

$$\begin{aligned} 3n + 3m + 2nm = 0 &\implies m = \frac{-3n}{2n+3} \\ &\implies 2m = \frac{-6n}{2n+3} = -3 + \frac{9}{2n+3} \\ &\implies 2m + 3 = \frac{9}{2n+3} \\ &\implies 2n+3 \text{ divides } 9 \text{ in } \mathbb{Z} \\ &\implies 2n+3 \in \{\pm 1, \pm 3, \pm 9\}. \end{aligned}$$

So, we get $(n, m) \in \{(-6, -2), (-3, -3), (-2, -6), (-1, 3), (0, 0), (3, -1)\}$.

3. Let $n, m \in \mathbb{Z} \setminus \{0\}$. We have

$$\begin{aligned} f(n) = f(m) &\iff 1 + \frac{2}{n} + \frac{3}{n^2} = 1 + \frac{2}{m} + \frac{3}{m^2} \\ &\iff \frac{2n+3}{n^2} = \frac{2m+3}{m^2} \\ &\iff 2m^2n + 3m^2 - 2mn^2 - 3n^2 = 0 \\ &\iff (m-n)(3n+3m+2nm) = 0 \end{aligned}$$

¹maximal set in sense of inclusion

$$\iff m = n \text{ or } 3n + 3m + 2nm = 0.$$

So we deduce from question 2. that for

$$(n, m) \in \{(-6, -2), (-2, -6), (-1, 3), (3, -1)\}$$

we have $f(n) = f(m)$, with $n \neq m$. Therefore the map f is not injective.

Or we can note directly that $f(-2) = f(-6)$ and $-2 \neq -6$. Then the map f is not injective.

4. From the previous question, it is easy to see that if we take $A := \mathbb{Z} \setminus \{-2, -1, 0\}$ (or $\mathbb{Z} \setminus \{-6, -1, 0\}$, $\mathbb{Z} \setminus \{-2, 0, 3\}$, $\mathbb{Z} \setminus \{-6, 0, 3\}$); the restriction $f|_A$ is injective and the set A is maximal.

Indeed, let $X \subset \mathbb{Z} \setminus \{0\}$ such that $f|_X$ is injective and $A \subset X$. We suppose that $-1 \in X$. Since $f|_X$ is injective then $3 \notin X$, which is not possible because $A \subset X$, so $-1 \notin X$. The same reasoning leads us to have $-2 \notin X$. Hence $X \subset A$, which means that $X = A$.

5. We have

$$\begin{aligned} B &= \{n \in \mathbb{Z} : |2n^2 + n + 4| \leq 2\} \\ &= \{n \in \mathbb{Z} : -2 \leq 2n^2 + n + 4 \leq 2\} \\ &= \{n \in \mathbb{Z} : 0 \leq 2n^2 + n + 6 \wedge 2n^2 + n + 2 \leq 0\} \\ &= \{-2, 1\}. \end{aligned}$$

So,

$$f(B) = \{f(-2), f(1)\} = \left\{\frac{3}{4}, 6\right\}$$

and

$$\begin{aligned} f^{-1}(B) &= \{n \in \mathbb{Z} \setminus \{0\} : f(n) \in B\} \\ &= \{n \in \mathbb{Z} \setminus \{0\} : f(n) = -2 \text{ or } f(n) = 1\} = \emptyset. \end{aligned}$$

Exercise 3 :

Let E and F be two nonempty sets and $f : E \rightarrow F$ a map. Show that

$$f \text{ is injective} \iff (\forall A, B \in \mathcal{P}(E), A \cap B = \emptyset \implies f(A) \cap f(B) = \emptyset).$$

Solution: (2+2)

$\implies$) Let $A, B \in \mathcal{P}(E)$ such that $A \cap B = \emptyset$. Assume that $f(A) \cap f(B) \neq \emptyset$. Let $y \in f(A) \cap f(B)$, so $\exists a \in A$ and $\exists b \in B$ such that $y = f(a)$ and $y = f(b)$. Thus, $f(a) = f(b)$ and the fact that f is injective give us $a = b$. Therefore, $a = b \in A \cap B$ which is impossible. So, $f(A) \cap f(B) = \emptyset$.

$\Leftarrow$) Let $x, x' \in E$ such that $x \neq x'$. So, $\{x\} \cap \{x'\} = \emptyset$. Thus, $f(\{x\}) \cap f(\{x'\}) = \emptyset$ (by hypothesis), i.e. $\{f(x)\} \cap \{f(x')\} = \emptyset$. Thus, $f(x) \neq f(x')$. Therefore, f is injective.